NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-27/68A
18 May 1977
SUPERSEDING
MIL-T-27/68
14 April 1976

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY TF5S21ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.

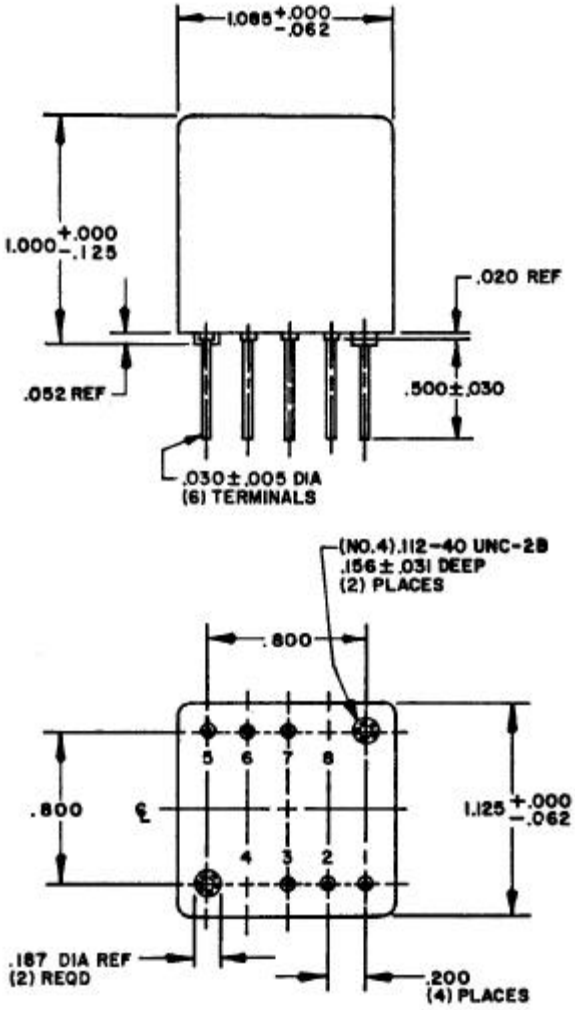
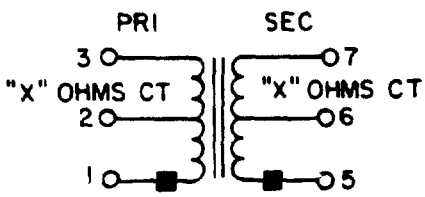


FIGURE 1. Dimensions and configuration.

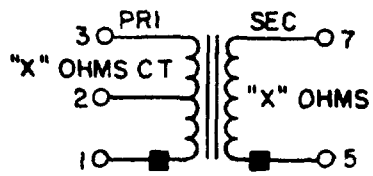
FSC 5950

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INCHES	MM	INCHES	MM
005	13	.187	4 75
.020	.51	.200	5.08
030	76	500	12 70
052	1 32	600	15 24
062	1 57	800	20.32
112	2 84	900	22 86
125	3 18	1 000	25 40
		1 085	27.56



A



B

WORKING VOLTAGE: "X" V
ALTITUDE . 70 kFT MAX
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAMS AND MARKING

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm
3 Unless otherwise specified, tolerance is ± 010 (25 mm)
4 Marking shall be on sides and/or top of case
5 Electrical values shall be marked as specified in table I, as applicable

FIGURE 1 Dimensions and configuration - Continued

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.
Power level: 500 milliwatts maximum at 100 hertz.
Working voltage 175 volts (except 355 volts on M27/68-07).
Frequency range 100 hertz to 8 kilohertz (except 100 hertz to 8 kilohertz on M27/68-01)

TABLE I. Electrical ratings.1/

Dash No.	Circuit diagram (see fig. 2)	Primary impedance ohms (1-3)	Secondary impedance ohms (5-7)	Primary DCR ohms ±25%	Secondary DCR ohms ±25%	Primary unbalanced DC current max mA
01	A	10k CT	10k CT	700	1,000	1.0
02	A	10k CT	600 CT	700	60	1.0
03	B	10k CT	4	700	0.4	1.0
04	B	1,500 CT	15	105	1.5	2.5
05	A	1,000 CT	600 CT	70	60	3.0
06	B	1,000 CT	4	70	0.4	3.0
07	A	600 CT	25k CT	42	6	4.0
08	A	600 CT	600 CT	42	2,600	4.0
09	B	600 CT	60	42	6	4.0
10	B	600 CT	15	42	1.5	4.0
11	B	600 CT	4	42	0.4	4.0
12	A	250 CT	600 CT	17	60	7.0

1/ Qualification testing and approval to M27/68-01 and all tests per table VII of MIL-T-27 for M27/68-07 shall be sufficient to grant qualification approval to M27/68-01 through M27/68-12.

Design and construction
Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case: Encapsulated.
Material: Plastic.
Terminals Pin (printed circuit).
Material Nickel iron alloy, tin plated.
Diameter 0.030 ±0.005 inch.
Length 0.500 ±0.030 inch.
Weight: 56.70 grams maximum.
Altitude 70,000 feet maximum.
Operating temperature range: -55°C to +130°C.

Terminal strength. MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:
At sea level: 500 volts rms (except 1,000 volts rms on M27/68-07).
At barometric pressure 300 volts rms (except 444 volts on M27/68-07).

Electrical characteristics See table II.
Frequency response
Reference frequency 1 kilohertz.
Polarity Additive, with terminals 3 and 5 connected.

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TABLF II Electrical characteristics.

Dash No.	Frequency response at 100 Hz to 10 kHz ±1 dB			
	Z_S (1-3) ohms	Z_L (5-7) ohms	E_L volts	Primary DC current (1-3) mA
01 1/	10 k	10 k	12.0	0.5
02	10 k	600	12.0	0 5
03	10 k	4	12.0	0 5
04	1,500	15	5.0	1.25
05	1,000	600	4.0	1 5
06	1,000	4	4.0	1.5
07	600	25 k	3.0	2.0
08	600	600	3.0	2.0
09	600	60	3.0	2.0
10	600	15	3.0	2.0
11	600	4	3.0	2.0
12	250	600	2.0	3.5

1/ Frequency response is 100 hertz to 8 kilohertz ±1 dB

Marking location: See figure 1.

Part number M27/68 - (dash number from table I)

NOTE Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - MU, WC, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity
Army - EL

Agent
DLA - ES

(Project 5950-0543-2)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-T-27768A TRANSFORMERS, AUDIO FREQUENCY, TF5S21ZZ

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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